

Data Analyst Capstone Project

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EXECUTIVE SUMMARY



Analyzed developer technology trends using survey, API, scraping and dashboard outputs.

Focused on current usage and future demand for languages, databases, platforms and web frameworks.

Python, JavaScript, SQL and TypeScript remain core skills across current and future demand.

PostgreSQL, SQLite and MongoDB show strong relevance in database trends.

Dashboard insights connect technology choices with demographics and workforce planning.



INTRODUCTION



Purpose: identify technology trends that matter for data-driven skill planning.

Target audience: recruiters, training teams, managers and junior data professionals.

Scope: programming languages, databases, platforms, web frameworks and demographics.

Value: converts raw survey and job data into actionable skill priorities.



METHODOLOGY



Data sources: Stack Overflow Developer Survey-style data, Jobs API output and web-scraped salary data.

Collection: API calls, CSV/Excel exports, web scraping and dashboard data assets.

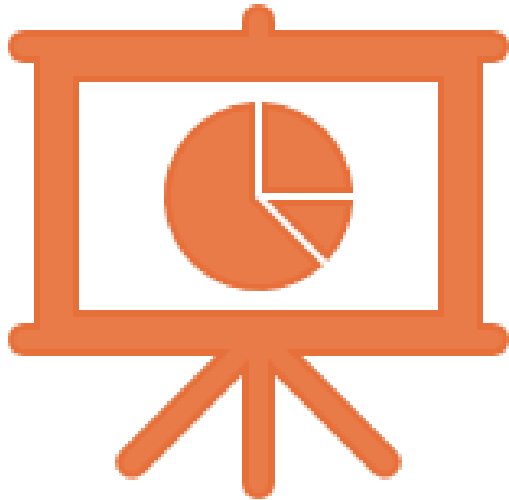
Wrangling: duplicate removal, missing-value checks, imputation and outlier review.

Analysis: Pandas, SQL and visual exploration of current vs. desired technologies.

Visualization: PowerPoint charts and dashboard-style summaries for storytelling.



Results

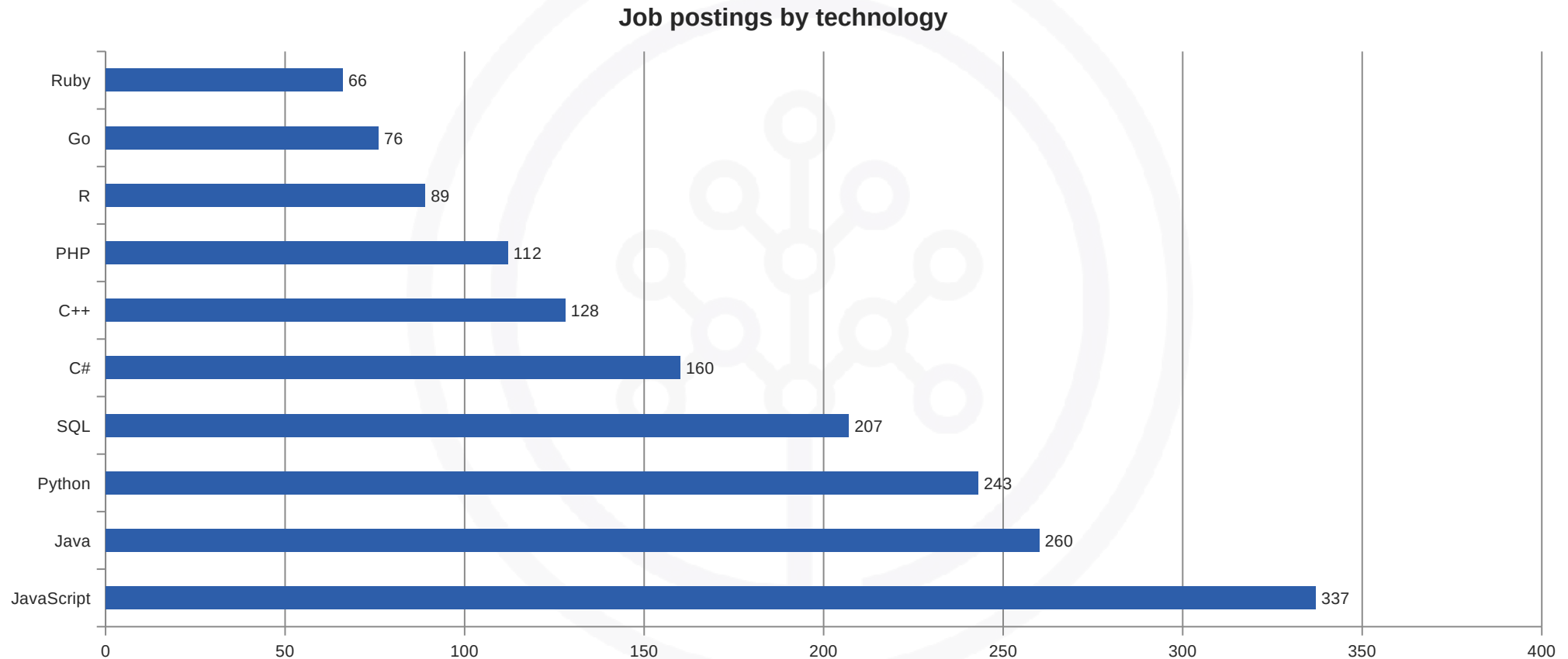


The next section presents the main analytical outputs used for the capstone story.

- Job-posting demand from API collection
- Salary comparison from web-scraped language data
- Current vs. future programming-language trends
- Current vs. future database trends
- Dashboard insights for technology usage, future demand and demographics

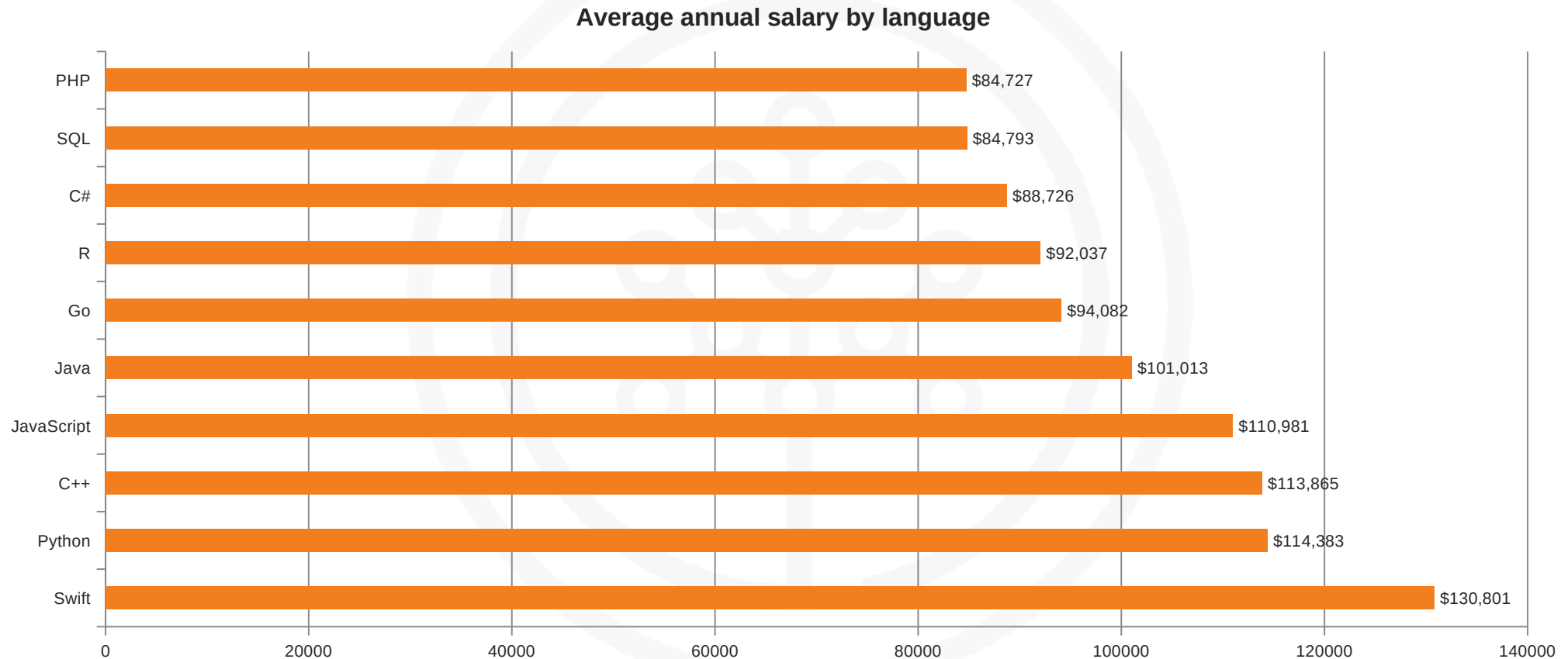
JOB POSTINGS

Job postings by programming skill, ordered from highest to lowest demand.



POPULAR LANGUAGES

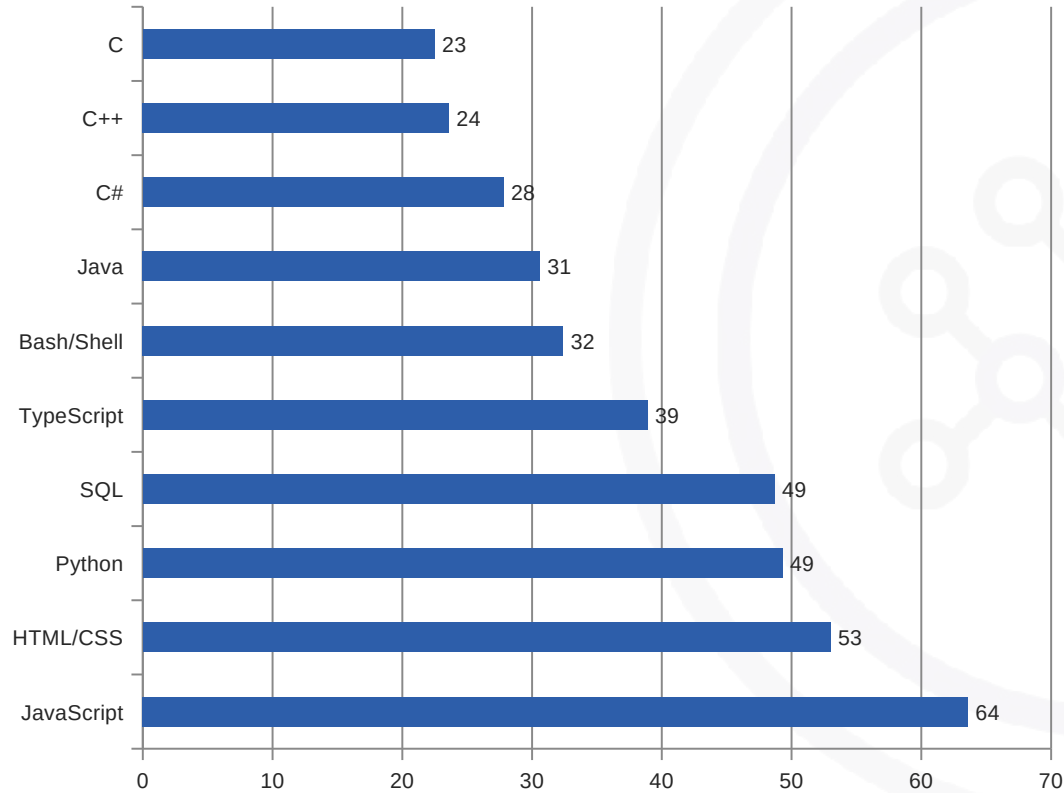
Average annual salary by programming language, ordered by salary.



PROGRAMMING LANGUAGES TRENDS

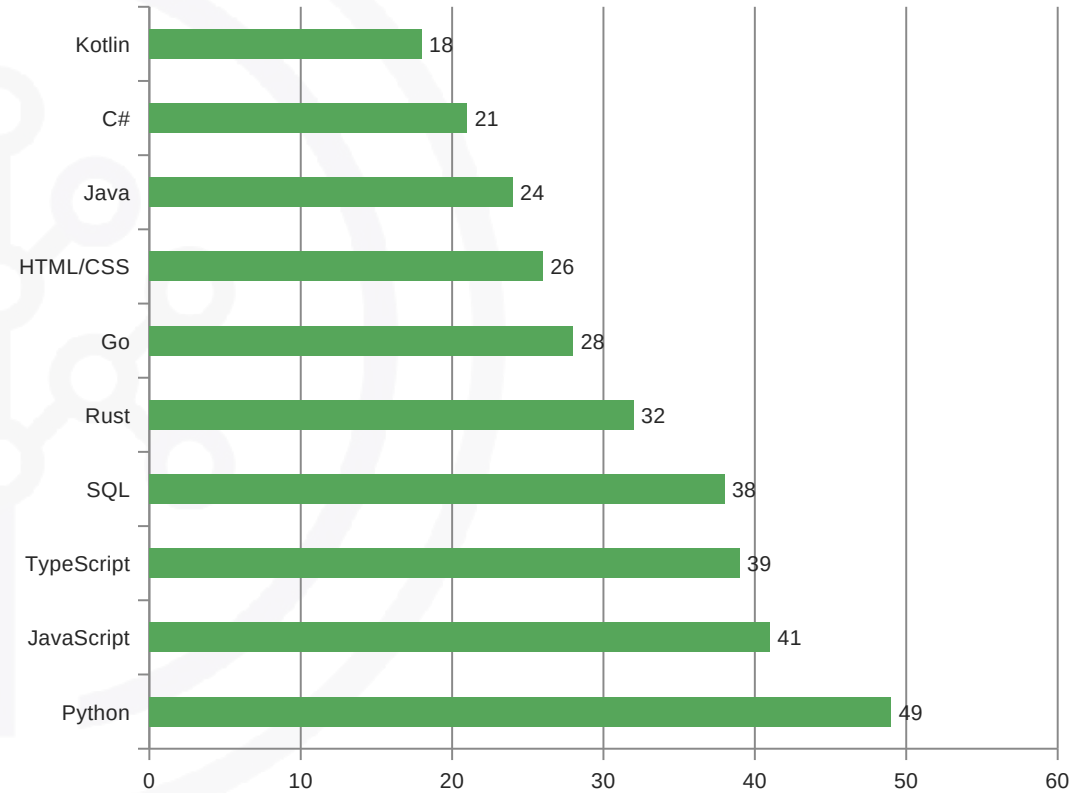
Current Year

Top 10 languages worked with (%)



Next Year

Top 10 languages desired (%)



Core web/data skills remain dominant, while Python, TypeScript, Rust and Go signal future growth.



PROGRAMMING LANGUAGES TRENDS - FINDINGS AND IMPLICATIONS

Findings

JavaScript, HTML/CSS, Python and SQL form the broadest current skill base.

Python remains highly attractive for future work, especially for data and automation roles.

TypeScript, Rust and Go indicate demand for scalable, reliable software development.

Implications

Entry-level analysts benefit from strong Python + SQL fundamentals.

Learning TypeScript/JavaScript improves collaboration with web/product teams.

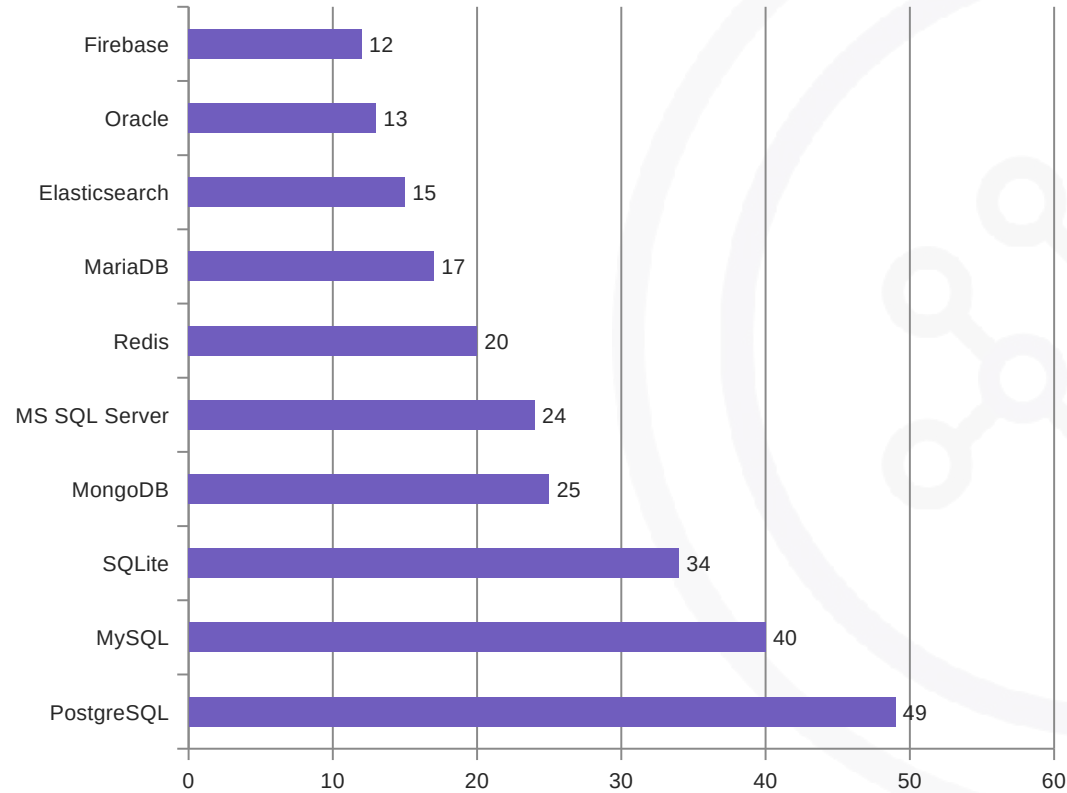
A practical portfolio should show both analysis skills and modern technical literacy.

DATABASE TRENDS

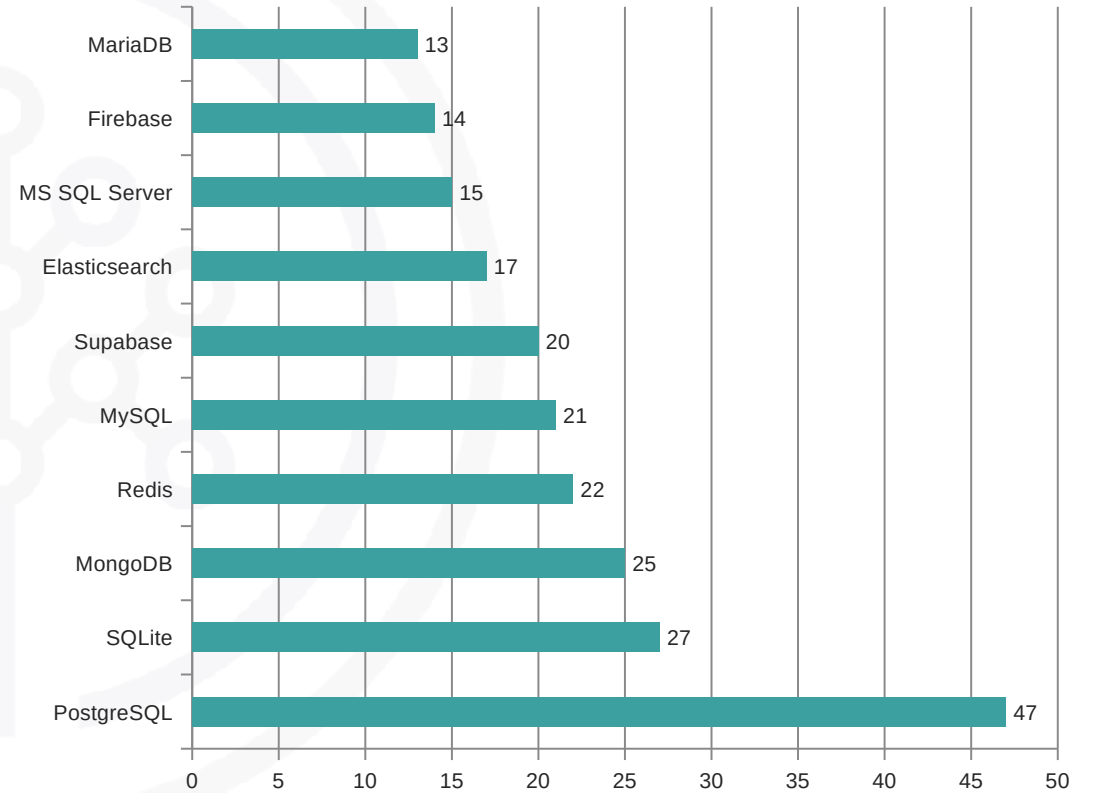
Current Year

Next Year

Top 10 databases currently used (%)



Top 10 databases desired (%)



DATABASE TRENDS - FINDINGS AND IMPLICATIONS

Findings

PostgreSQL leads both current usage and future interest.

MySQL and SQLite remain important due to broad adoption and ease of use.

MongoDB, Redis and Supabase reflect demand for flexible and cloud-oriented data stacks.

Implications

SQL skills remain a core requirement for data analysis roles.

PostgreSQL is a high-value database to practice in portfolio projects.

Exposure to NoSQL/cache/cloud databases strengthens a modern analyst profile.

DASHBOARD

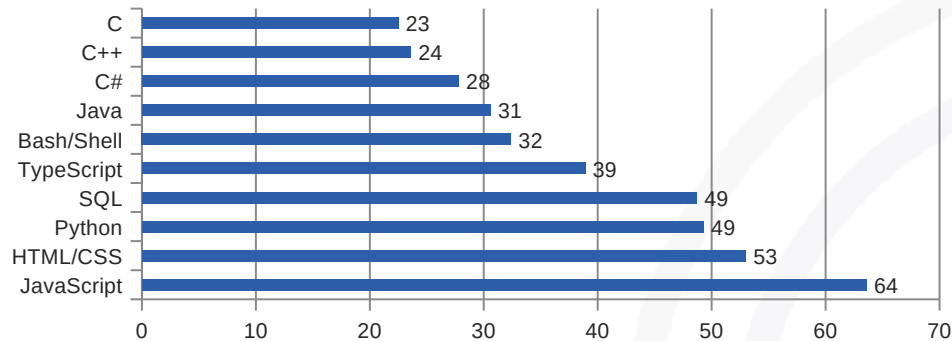


The dashboard section summarizes three views: current usage, future demand and demographics.

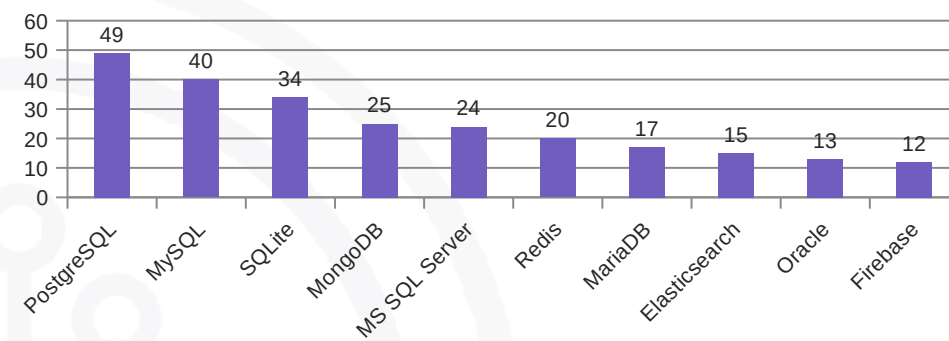
Tab 1: Current Technology Usage
Tab 2: Future Technology Trends
Tab 3: Demographics

DASHBOARD TAB 1: Current Technology Usage

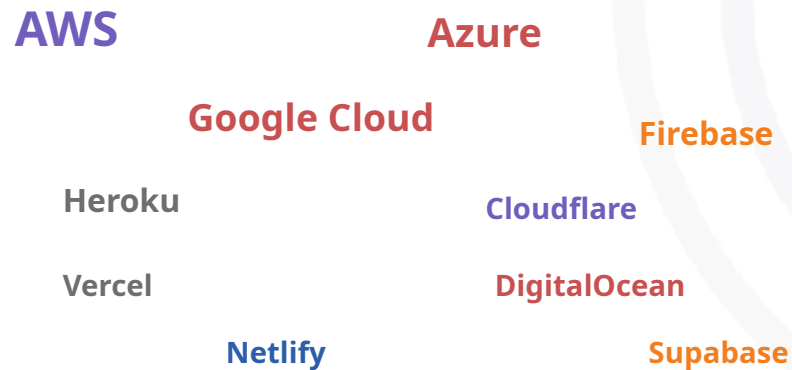
Top 10 languages used



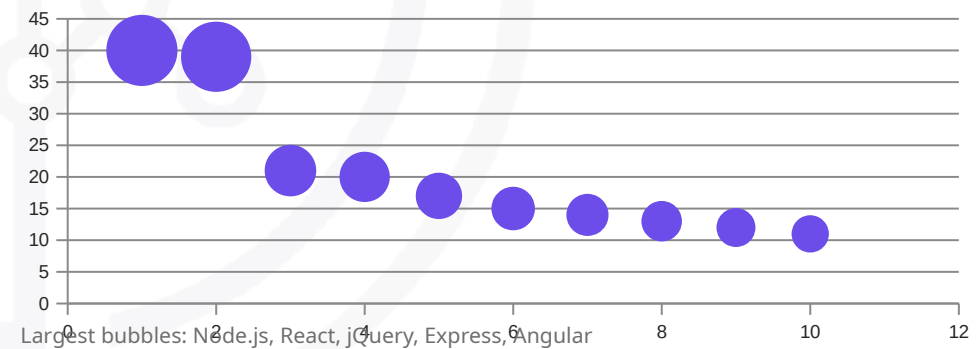
Top 10 databases used



Top platforms used

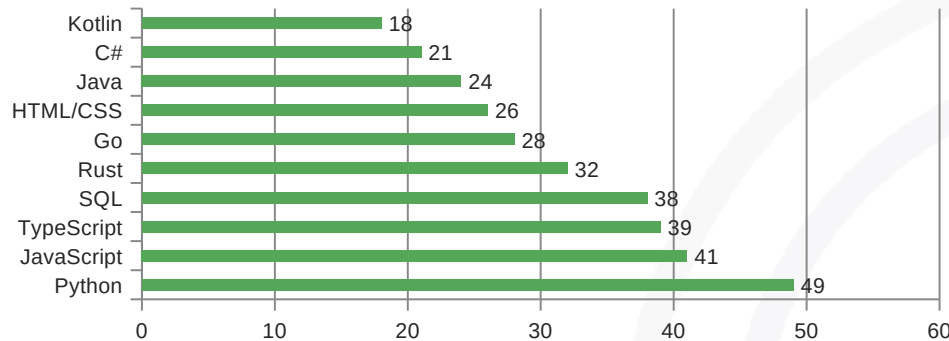


Top web frameworks used

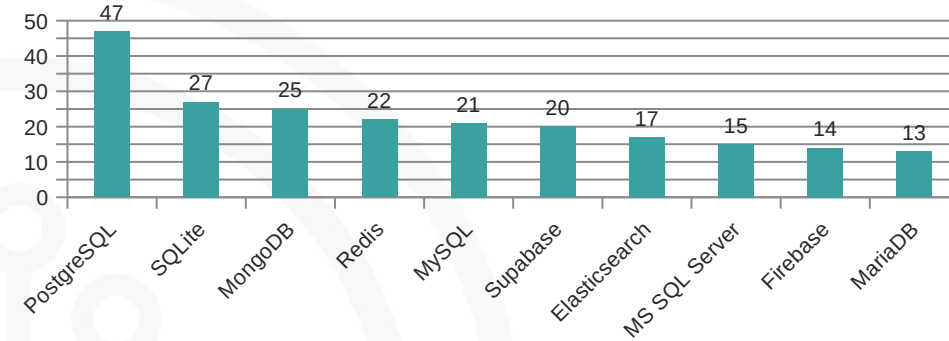


DASHBOARD TAB 2: Future Technology Trends

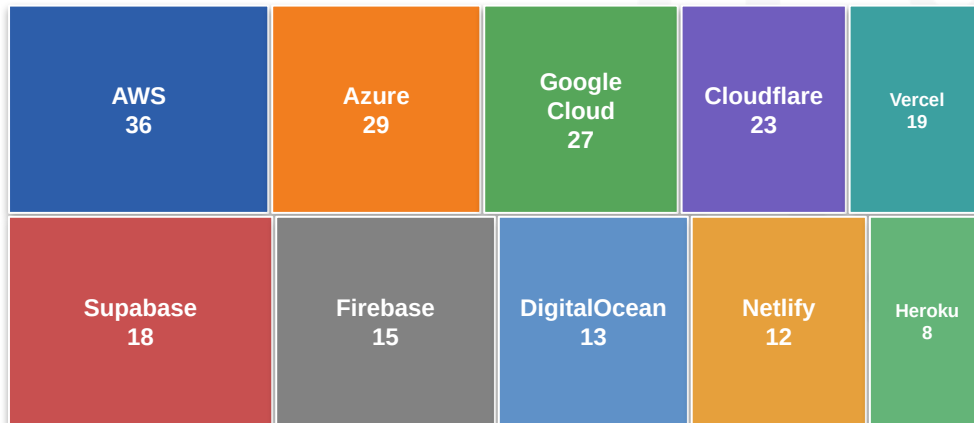
Top 10 languages wanted



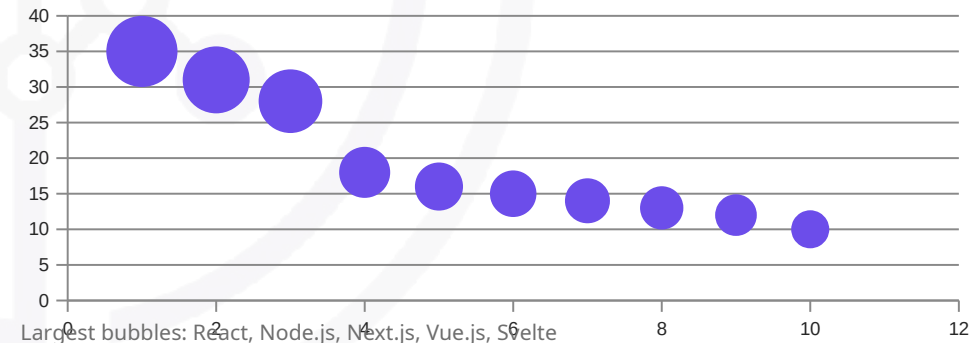
Top 10 databases wanted



Top platforms wanted

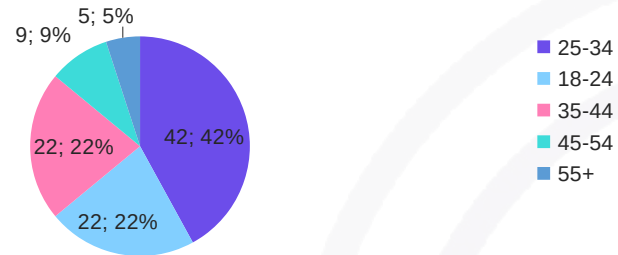


Top web frameworks wanted

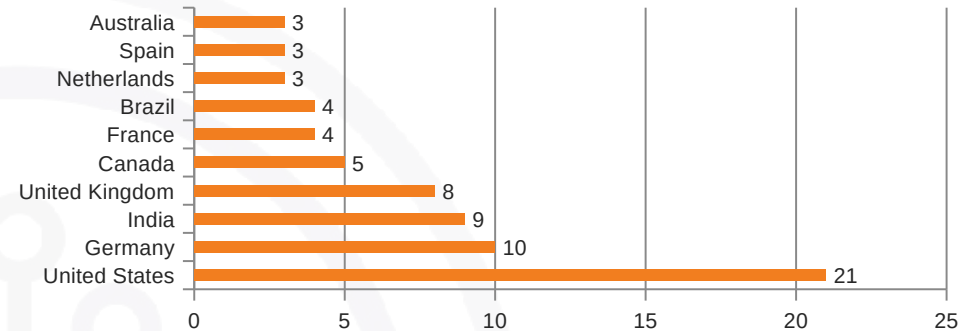


DASHBOARD TAB 3: Demographics

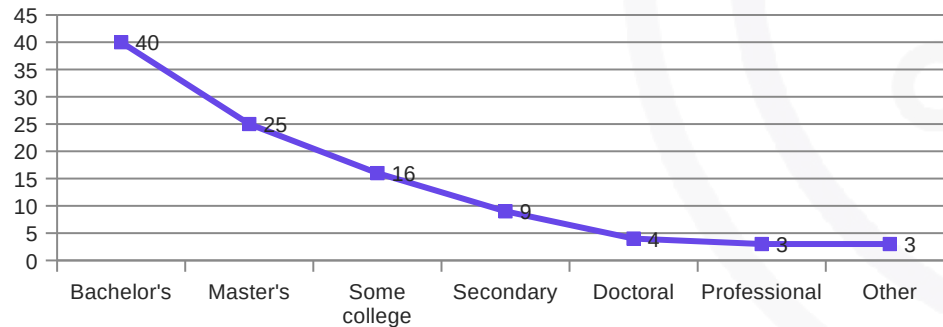
Respondent distribution by age



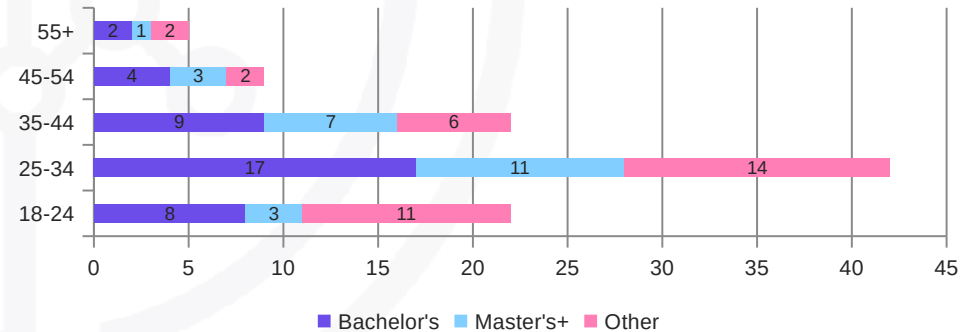
Respondent count by country



Respondent distribution by education



Age by education level



DISCUSSION



The dashboard connects technology usage with future demand and respondent profiles.

Current usage is strongest around established languages and databases, meaning core fundamentals remain valuable.

Future demand shows interest in technologies that support automation, cloud-native development and modern web applications.

Demographics suggest the results reflect a broad developer population, not a single experience group.

OVERALL FINDINGS AND IMPLICATIONS

Findings

Programming and database skills remain the strongest signal across the analysis. Python, SQL and JavaScript/TypeScript form a practical core for data and product teams. PostgreSQL and cloud platforms indicate where data workflows are moving.

Implications

Learners should prioritize SQL, Python, Power BI/Cognos-style dashboarding and portfolio projects. Employers should align training programs with current tools and future demand signals. Analysts who can combine data cleaning, visualization and business storytelling create more value.



CONCLUSION



The project demonstrates an end-to-end data analysis workflow: collection, wrangling, analysis, visualization and reporting.

Technology trends confirm the continuing importance of Python, SQL, JavaScript/TypeScript and relational databases. Dashboards help transform detailed survey data into clear decisions about skill priorities.

The final recommendation is to focus learning and portfolio work on practical, business-facing data analysis skills.



APPENDIX

Appendix and source note

Supporting analyses include job postings and programming-language salary comparisons.

Technology trend slides compare current usage with future technology interest.

Dashboard slides summarize Current Technology Usage, Future Technology Trends, and Demographics.

These supporting views reinforce the findings presented in the report.

